

School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing
Project Scribe Submission

Adaptive Traffic Control Using Las Vegas Algorithm

Group Number: S1.G7.ITS

Date: March 10th, 2026

Scribe Question 1: Project System and Objective

What is the probabilistic problem being addressed in your project? Clearly state the system objective and identify the primary sources of uncertainty involved.

Answer:

Scribe Question 2: Key Random Variables and Uncertainty Modeling

Identify the key random variables in your project and describe how uncertainty is modeled for each. Clearly state any probabilistic assumptions currently being made.

Answer:

Scribe Question 3: Probabilistic Reasoning and Dependencies

Explain how probabilistic relationships (e.g., dependence, independence, conditionality) are used within your project system to support reasoning, inference, or decision-making.

Answer:

Scribe Question 4: Model–Implementation Alignment

Describe how the current probabilistic model aligns with the project's implementation or experimental setup. Highlight any assumptions that influence design or evaluation choices.

Answer: We run the process as the end-to-end pipeline between SUMO, the microscopic simulator that creates the random traffic states, and our Python code, which produces the deterministic baseline algorithm of our backend implementation. The TraCI interface is then used to interface with the two, transmitting the traffic state to Python and returning control signals went to SUMO.

The first assumption that affects our set up is that we are able to treat stochastic inputs as fixed values to execute fixed mathematical formulas. This is correspondingly reflected in our Python code that computes

the time of cycles analytically with no randomized control decisions, relying heavily on the Webster ($C_o = \frac{1.5L+5}{1-\sum y_i}$) and Modified Webster ($C_o = \frac{1.978L+5.109}{1-0.9013\sum y_i}$) equations.

Scribe Question 5: Cross-Milestone Consistency and Change

Describe the current state of your project's probabilistic model. Identify the assumptions, components, or relationships that are presently well-defined, and indicate which aspects are expected to evolve or require refinement in subsequent milestones. Briefly justify your reasoning.

Answer: In this milestone, we have managed to carry out a deterministic baseline strategy. The existing specifications that are currently clear are the mathematical calculation of the optimal cycle and the proportional distribution of green time according to the crucial lane flow rates ($y_i = \frac{\max(\epsilon_s, \epsilon_l, \epsilon_{sr})}{s}$). This baseline gives us a stable reference for performance comparison.

But since this deterministic algorithm requires correct flow estimation and relies on a set formula structure then it is incapable of predicting and cannot adjust according to long-term trends. Our decision making elements will change in later milestones. In order to deal with very non-stationary traffic, we will switch to probabilistic cycle optimization and Deep traffic state prediction.

Scribe Question 6: Open Issues and Responsibility Attribution

Identify any unresolved probabilistic questions or ambiguities in the project at this stage, and indicate which role(s) or tasks are responsible for addressing them in the next milestone.

Answer: The unanswered probabilistic questions at this stage are primarily on the weaknesses of the system in bad performance situations. Due to the reliance of our present deterministic baseline on proper estimation of traffic flow, and the fact that it relies on a fixed formula structure there is uncertainty in how the system will operate in cases of very non-stationary traffic, incident-induced congestion, excessive imbalance in demand, or sensor errors. The existing model has not much predictive capacity and fails to comprehend long-term trends.

To overcome these deficiencies of the following milestone, the activities of our team are aimed at shifting towards the non-deterministic optimization. Our future directions will be to investigate our outlined future directions, which are to build probabilistic cycle optimization and deep traffic state prediction. In addition, work will include exploring more of signal policies in Reinforcement Learning and multi-intersection coordination as a way of managing the complex, stochastic characteristics of real-world traffic.

End of Submission